

SMART CANTEEN

QUEUE MANAGEMENT SYSTEM

Detailed Technical Project Report

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PROJECT FOCUS Digital canteen ordering and queue management using structured order records, token generation, FIFO queue processing, status tracking, and file-based record handling in C.

Executive Snapshot

The Smart Canteen Queue Management System is designed to organize the college canteen ordering process and reduce the difficulties caused by long queues, waiting time, counter crowding, and poor order management. The solution models the canteen as a controlled order-processing system in which students place orders, receive tokens, and wait for their orders to progress through defined status stages. Canteen staff manage the queue in sequence and update order status until the food is ready.

Dimension	System Approach
Users	Students and canteen staff
Menu	Display and manage available food items
Ordering	Select food items and place an order
Token	Generate a unique token number for every order
Queue	Process orders in First-In-First-Out (FIFO) sequence
Status	Track Pending, Preparing, and Ready states
Records	Maintain structured order information using C data structures and file handling

01 Project Overview & Problem Definition

Abstract

The Smart Canteen Queue Management System is a C-based project intended to improve the efficiency of food ordering and queue handling in a college canteen. In a traditional setup, students may stand in long queues, spend valuable break time waiting, and gather around the counter while staff manually manage multiple orders. The proposed system introduces an organized flow for menu viewing, order placement, token generation, queue processing, and status updates.

The system identifies students and canteen staff as the primary users. Students can select available food items and place orders, while staff manage the incoming queue and update the state of each order. Orders are processed according to FIFO sequence, and each order can move through the states Pending, Preparing, and Ready. The overall goal is a faster, clearer, and more manageable canteen experience.

1. The Canteen Queue Management Problem

The main problem is the absence of a systematic mechanism for handling a high volume of student orders during peak canteen periods. Manual ordering can result in long queues, crowding at the service counter, difficulty in maintaining order sequence, and uncertainty about when an order will be ready.

CORE FAILURE Long queues, time wastage, crowding, and weak order management can reduce the efficiency of canteen operations and lead to student dissatisfaction.

Impact

1. Students may spend a significant part of their break waiting for food.
2. Crowding can increase around the ordering and collection counter during peak hours.
3. Manual handling makes it harder to maintain a clear sequence of orders.
4. Students may not know whether an order is pending, being prepared, or ready.
5. Unorganized order handling can reduce both staff efficiency and student satisfaction.

02 Objectives, Requirements & Technology Stack

2. Project Objectives

1. Reduce the waiting time experienced by students at the canteen.
2. Manage food orders through an organized digital queue.
3. Simplify the food-ordering process for students.
4. Generate a unique token number for each order.
5. Maintain the order sequence using First-In-First-Out (FIFO) logic.
6. Allow canteen staff to update order status as the order progresses.
7. Improve the overall student canteen experience.
8. Help canteen staff manage orders and available resources more efficiently.

3. Requirements

Requirement	Responsibility
User Management	Identify and support students and canteen staff
Food Menu	Add and display available food items
Order System	Allow students to select items and place orders
Token System	Generate a unique token number for each order
Queue Management	Process all orders in correct FIFO sequence
Order Status	Track Pending, Preparing, and Ready states
Record Handling	Store and retrieve order information when required

4. Technology Stack

Area	Technology	Role
Language	C (C11 / GNU11)	Core implementation
Data Structures	Structures, arrays, queue-oriented records	Store and manage orders
Functions	Modular C functions	Separate major system operations
Decision Making	if / else / switch	Control menus and system decisions
Loops	for / while	Repeated processing
Pointers	C pointers	Reference and efficient record handling
File Handling	C file I/O	Store and retrieve records
Environment	GCC / standard C environment	Compile and execute the application

03 System Design & Queue Processing

5. System Architecture

The system is organized as a sequence of functional layers. The student interaction layer captures the order request; the menu and order layer validates the selected items; the token layer assigns a unique token; the queue layer maintains FIFO order; and the staff layer processes and updates orders until completion.

Layer	Components	Responsibility
User Layer	Students / Canteen Staff	Interact with the system
Menu Layer	Food items and prices	Display available items
Order Layer	Order records	Capture and validate orders
Token Layer	Token generator	Assign a unique token
Queue Layer	FIFO queue	Maintain processing sequence
Status Layer	Pending / Preparing / Ready	Track order progress
Record Layer	Structures / files	Maintain order information

6. Workflow Design

END-TO-END WORKFLOW Student selects food → order created → token generated → order enters FIFO queue → staff prepares order → status updated → order marked Ready → student collects order.

7. Queue Processing Logic

The queue is based on First-In-First-Out (FIFO) processing. The earliest valid order added to the queue is served before later orders. This provides predictable ordering and reduces confusion when many students place orders close together.

Step	Operation	Result
1	Receive order	Student order data is captured
2	Validate selection	Selected menu items are checked
3	Generate token	Unique token assigned
4	Enqueue	Order added at the rear of the queue
5	Process front order	Oldest waiting order is prepared
6	Update status	Pending → Preparing → Ready
7	Complete	Processed order removed or archived

8. Data Representation

A structured order record can contain fields such as token number, student identifier, selected item information, quantity, total amount, and current status. Structures provide a natural way to group these related attributes into a single order object that can be passed between functions.

04 Implementation & Core C Concepts

9. Modular Program Design

The implementation can be divided into independent functions so that each operation is easier to test and maintain. Typical modules include menu display, order creation, token generation, enqueue operation, dequeue operation, order-status update, record display, and file handling.

Module	Purpose
displayMenu()	Display available food items
placeOrder()	Accept student selections and create an order
generateToken()	Create the next unique token number
enqueueOrder()	Add a new order to the queue
processOrder()	Handle the order at the front of the queue
updateStatus()	Change the state of an order
displayOrders()	Show active order records
saveRecords() / loadRecords()	Persist records using file handling

10. C Programming Concepts Demonstrated

1. Variables and data types for representing tokens, quantities, prices, and menu values.
2. Decision-making constructs for menu selection and order validation.
3. Loops for displaying menus and processing multiple orders.
4. Functions for modular and reusable program design.
5. Structures for representing menu items and complete order records.
6. Pointers for passing and manipulating structured records efficiently.
7. Arrays or queue-oriented storage for maintaining multiple orders.
8. File handling for saving and retrieving persistent order information.

11. Token Generation

A token number acts as the visible identifier for an order. Each accepted order receives the next available token so that students and staff can refer to a specific order without ambiguity. Token generation also supports the FIFO queue because the sequence of tokens reflects the sequence in which orders are accepted.

DESIGN RULE A token should be assigned only after the order has been validated and accepted into the system.

05 Order Management, Status Tracking & File Handling

12. Order Status Lifecycle

Each order follows a simple status lifecycle. Immediately after placement, the order is marked Pending. When the canteen starts preparing the order, the status changes to Preparing. Once the food is ready for collection, the order becomes Ready.

Status	Meaning	Typical Action
Pending	Order accepted and waiting in the queue	Wait for processing
Preparing	Order is currently being prepared	Canteen staff work on the order
Ready	Food is prepared and can be collected	Student collects the order

13. Record Management

Order records must remain consistent while the system is processing multiple transactions. A structured record approach allows the program to keep related values together and makes operations such as search, display, update, and archival easier to implement.

14. File Handling

File handling can be used to maintain a persistent record of menu items, order information, or completed transactions. The program can write records to a file when an order is placed or completed and read them back when the application needs to restore or display stored information.

Operation	C Mechanism	Purpose
Open	fopen()	Open a file in the required mode
Write	fprintf() / fwrite()	Store order or menu data
Read	fscanf() / fread() / fgets()	Retrieve stored information
Append	a / ab mode	Add new records without removing earlier ones
Close	fclose()	Safely finish file operations

15. Search and Display

The system can search order records using a token number or another identifying field. Display functions can then present current orders, their status, and the relevant menu or transaction information to students or staff.

06 Testing, Results & Validation

16. Validation Strategy

Test	Expected Safe Behavior	Purpose
Menu Display Test	Available food items are displayed correctly	Validate menu module
Order Creation Test	Valid order is stored successfully	Validate order input
Token Test	Each accepted order receives a unique token	Validate token generation
FIFO Queue Test	Oldest order is processed first	Validate queue behavior
Status Update Test	Order moves through valid status states	Validate tracking
File Save / Load Test	Records are stored and retrieved correctly	Validate persistence
Invalid Input Test	Incorrect choice is rejected safely	Validate input handling
Multiple Order Test	Several orders remain in correct sequence	Validate normal peak usage

17. Expected Results

1. Reduced confusion at the order counter through token-based identification.
2. Consistent order processing through FIFO queue management.
3. Clear visibility of order progress using status values.
4. Better organization of student orders for canteen staff.
5. More efficient use of the available canteen service time.

6. Availability of stored records for later reference when file handling is enabled.

18. Output Summary

A successful execution should demonstrate a complete flow from menu selection to order completion. The student places an order, receives a token, waits in the queue, and can identify the current state of the order. At the staff side, orders are processed in sequence and their status is updated until the order becomes Ready.

07 Benefits, Limitations & Future Enhancements

19. Key Benefits

Benefit	Outcome
Reduced waiting time	Students spend less time in unorganized queues
Organized queue	Orders follow a clear FIFO sequence
Improved order visibility	Students and staff can track status
Better operations	Staff can process orders systematically
Improved experience	Less crowding and confusion at the counter
Data and records	Stored information can support later review and reporting

20. Limitations

1. The basic C application is primarily intended as an academic prototype rather than a production-scale distributed application.
2. Actual online ordering would require a user interface and a network or backend service beyond the core C program.
3. Persistent storage through simple files has fewer capabilities than a dedicated database system.
4. The system depends on correctly entered menu, student, and order information.
5. Peak-volume performance would need to be measured with a larger test dataset and realistic concurrent usage.

21. Future Enhancements

1. Add a graphical or web-based interface for students and staff.
2. Connect the application to a database for scalable and reliable storage.
3. Provide online order status notifications and digital token display.
4. Add authentication for students and canteen staff.
5. Generate reports for peak hours, popular items, and completed orders.
6. Extend the design to support multiple counters or multiple canteen sections.

FUTURE DIRECTION The queue-management core can be extended into a larger campus food-service platform while retaining the same fundamental concepts of tokens, FIFO processing, status tracking, and structured record management.

08 Conclusion & Project Keywords

22. Conclusion

The Smart Canteen Queue Management System presents a practical application of C programming to a common campus problem. By replacing an unorganized manual queue with a structured order and token workflow, the system provides a

clearer and more predictable way to manage food orders. The use of FIFO queue processing ensures that accepted orders are handled in sequence, while status tracking gives a simple view of order progress from Pending to Preparing to Ready.

From a programming perspective, the project provides opportunities to apply core C concepts such as structures, functions, arrays, pointers, decision-making, loops, and file handling in a single integrated problem. The result is an academic prototype that demonstrates how fundamental programming concepts can be combined to design a useful real-world system.

PROJECT IN ONE SENTENCE The Smart Canteen Queue Management System uses C-based order records, token generation, FIFO processing, and status tracking to create a more organized and efficient college canteen workflow.

23. Keywords

C Programming • Structures • Functions • Arrays • Pointers • File Handling • Queue • FIFO • Token Generation • Order Management • Status Tracking • Menu Management • Canteen Automation

09 References

24. References

1. Kernighan, B. W., and Ritchie, D. M. The C Programming Language, 2nd Edition, Prentice Hall.
2. ISO/IEC 9899:2018. Programming Languages — C. International Organization for Standardization.
3. GNU Project. The GNU C Library Reference Manual. Documentation for standard C library facilities and file I/O.
4. Course material for CSA-0211 – C Programming, including operators, decision-making, looping, arrays, strings, functions, pointers, and file handling.
5. Project-specific requirements and workflow derived from the provided Smart Canteen Queue Management System reference material.

25. Source Project Information

Project title: Smart Canteen Queue Management System. The supplied project material defines the problem, objectives, users, food menu, order system, token system, FIFO queue management, order-status tracking, system design, deployment, and benefits used as the basis for this report.